For example, in an open interval (4, 7), 4 and 7 are not elements of this interval.

But, 4.001 and 6.99 are elements of this interval.

(ii) Closed interval

The set $\{x : a \leq x \leq b\}$ is a closed interval and is denoted by $[a, b]$.

In the interval $[a, b]$, the boundary points a and b are included.

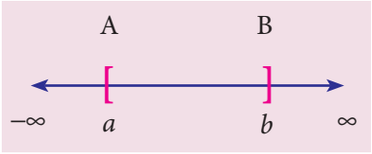


Fig. 5.3

For example, in an interval [4, 7], 4 and 7 are also elements of this interval.

Also we can make a mention about semi closed or semi open intervals.

$(a, b] = \{x : a < x \leq b\}$ is called left open interval and

$[a, b) = \{x : a \leq x < b\}$ is called right open interval.

In all the above cases, $b - a = h$ is called the length of the interval.

5.1.5 Neighbourhood of a point

Let 'a' be any real number. Let $\varepsilon > 0$ be arbitrarily small real number.

Then $(a - \varepsilon, a + \varepsilon)$ is called an ε - neighbourhood of the point 'a' and denoted by $N_{a, \varepsilon}$

For examples,

$$N_{5,\frac{1}{4}} = \left(5 - \frac{1}{4}, 5 + \frac{1}{4}\right) = \left\{x : \frac{19}{4} < x < \frac{21}{4}\right\}$$

$$N_{2,\frac{1}{7}} = \left(2 - \frac{1}{7}, 2 + \frac{1}{7}\right) = \left\{x : \frac{13}{7} < x < \frac{15}{7}\right\}$$

5.1.6 Function

Let X and Y be two non-empty sets of real numbers. If there exists a rule f which associates to every element $x \in X$, a unique element $y \in Y$, then such a rule f is called a function (or mapping) from the set X to the set Y. We write $f : X \longrightarrow Y$.

DO YOU KNOW?

The function notation $y = f(x)$ was first used by Leonhard Euler in 1734 - 1735

The set X is called the domain of f, Y is called the co-domain of f and the range of f is defined as $f(X) = \{f(x) / x \in X\}$. Clearly $f(X) \subseteq Y$.

A function of x is generally denoted by the symbol $f(x)$, and read as *"function of x" or "f of x"*.

5.1.7 Classification of functions

Functions can be classified into two groups.

(i) Algebraic functions

Algebraic functions are algebraic expressions using a finite number of terms, involving only the algebraic operations addition, subtraction, multiplication, division and raising to a fractional power.

Examples : $y = 3x^4 - x^3 + 5x^2 - 7$, $y = \dfrac{2x^3 + 7x - 3}{x^3 + x^2 + 1}$, $y = \sqrt{3x^2 + 6x - 1}$ are algebraic functions.

(*a*) $y = 3x^4 - x^3 + 5x^2 - 7$ is a polynomial or rational integral function.

(*b*) $y = \dfrac{2x^3 + 7x - 3}{x^3 + x^2 + 1}$ is a rational function.

(*c*) $y = \sqrt{3x^2 + 6x - 1}$ is an irrational function.

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